

Project Demonstration

Overview

The Project Demonstration showcases the complete functionality of OptiCrop by presenting how users interact with the system from registration to crop prediction. The demonstration highlights the integration of the frontend, backend, Machine Learning model, and database, proving that the application operates as a complete and functional smart agriculture solution.

Demonstration Objectives

The project demonstration aims to:

- Present the complete working of OptiCrop.
 - Show the interaction between different system modules.
 - Demonstrate accurate crop prediction.
 - Verify smooth communication between the frontend, backend, and database.
 - Highlight the practical benefits of the application.
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Demonstration Environment

The project was demonstrated using the following environment:

Component	Technology
Frontend	React.js
Backend	Flask
Database	MongoDB
Machine Learning	Scikit-learn
Development Tool	Visual Studio Code
Version Control	Git & GitHub

Demonstration Workflow

The demonstration follows the complete user journey through the application.

Step 1 – User Login

The user logs into the application using valid credentials. New users can register before accessing the system.

Step 2 – Enter Agricultural Parameters

The user enters the required values:

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)
- Temperature
- Humidity
- Soil pH
- Rainfall

The application validates the inputs before processing them.

Step 3 – Crop Prediction

After submitting the data, the Machine Learning model analyzes the inputs and predicts the most suitable crop based on the trained dataset.

Step 4 – Display Result

The application displays:

- Recommended Crop
- Prediction Summary
- Entered Input Values

The result is presented in a clear and easy-to-understand format.

Step 5 – Save Prediction

The generated prediction is stored in the MongoDB database, allowing users to access their prediction history later.

Step 6 – Submit Feedback

Users can provide feedback about the prediction, helping improve future versions of the application.

Key Features Demonstrated

The demonstration includes:

- Secure User Authentication
 - Responsive User Interface
 - AI-Based Crop Recommendation
 - Prediction History
 - Feedback Collection
 - Database Integration
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Demonstration Results

The demonstration confirmed that:

- User authentication functions correctly.
 - Agricultural inputs are validated successfully.
 - Crop recommendations are generated accurately.
 - Prediction records are stored without errors.
 - The application responds efficiently under normal usage.
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Screenshots

Include screenshots of the following pages in this section:

- Login Page
- Registration Page
- Home Dashboard
- Crop Recommendation Form
- Prediction Result
- Prediction History
- Feedback Page

Example:

![Login Page](images/login.png)

![Dashboard](images/dashboard.png)

![Crop Prediction](images/prediction.png)

![Prediction Result](images/result.png)

Conclusion

The project demonstration validates that OptiCrop successfully integrates Artificial Intelligence, Machine Learning, and modern web technologies to deliver an intelligent crop recommendation system. The application performs all core functionalities as planned and demonstrates its ability to support data-driven agricultural decision-making through a reliable, user-friendly, and scalable platform.